



# **Raspberry PI Setup**

## Physical setup

The Raspberry Pi is powered via Power over Ethernet (PoE) from the network switch. It is also connected to the Elgato Stream Deck using a USB cable from the Raspberry Pi to the Stream Deck.

## Introduction

The Raspberry Pi should remain powered on and hosted 24/7. If it goes offline, there are steps available to bring it back online.

The Raspberry Pi is used to host the Elgato Stream Deck so that the Stream Deck does not require a dedicated PC to operate. With this setup, the computer does not need to run the Elgato Stream Deck Software or Bitfocus Companion. Instead, Companion runs directly on the Raspberry Pi and handles all Stream Deck functions.

## Getting Started

The Raspberry Pi is running Companion OS as its operating system. Unlike Microsoft Windows or macOS, Companion OS primarily uses a command-line interface rather than a traditional graphical desktop environment.

If you would like to configure the Elgato Stream Deck or create a new button, [go here](#):

If the Raspberry Pi is offline or you would like to view system information or manage the Pi, [go here](#).

## Stream Deck Information

The Elgato Stream Deck configuration dashboard hosted on the Raspberry Pi can be accessed at 192.168.20.100:8000 on a traditional web browser. When connecting, make sure to include the port number (8000) in the address. After connecting, you will be prompted to enter a password to access the dashboard. Password: **streamER37**. Only access the dashboard if you know what you are doing.

Once inside the dashboard, you should primarily work within the “Buttons” tab located on the left side of the interface. Avoid modifying other sections unless necessary. Please note that creating a new page may disrupt the existing programming, since many of the current functions rely on the established page layout. If you need to create a new page, add it as the last page to avoid breaking existing configurations. To learn more about using the dashboard, [refer to the official documentation for Bitfocus Companion](#).

## Setting up the Raspberry PI to remote into it

There are two ways to connect to the Raspberry Pi.

### Connect via SSH (Recommended)

You can remotely access the Raspberry Pi using SSH from any computer connected to the campus network. Use the following command in a terminal: **ssh admin@192.168.20.100**. You will then be prompted to enter a password. Password: **Admin**. Make sure the “A” is capitalized, as the password is case-sensitive.

### Connect Directly with a Monitor and Keyboard

You can also access the Raspberry Pi directly through its command interface using a monitor and keyboard.

### To do this, you will need:

- A micro HDMI to HDMI cable
- A monitor
- A keyboard

Plug the micro HDMI end of the cable into the rightmost micro HDMI port on the Raspberry Pi (the one closest to the USB-C power port). Connect the standard HDMI end of the cable to the monitor. Plug the keyboard into one of the USB ports on the Raspberry Pi. Once everything is connected, the command interface will appear on the monitor, allowing you to interact directly with the system.

## Troubleshoot Raspberry PI Interface

Once the monitor is connected to the correct input on the Raspberry Pi, the command line interface should appear on the screen. You will be prompted to log in with the following credentials:

- Username: **admin**
- Password: **Admin** (make sure the “A” is capitalized)

### If the command line does not appear, try the following steps:

- Check the monitor input and make sure it is set to the correct HDMI source.
- Verify the cable connection. The micro HDMI cable must be plugged into the rightmost HDMI port on the Raspberry Pi (the one closest to the USB-C power port).
- If the cable is plugged into the left micro HDMI port, the display may not appear.
- If the display still does not show, power cycle the Raspberry Pi by unplugging the power cable (PEO), waiting 5 seconds, and then plugging it back in. Please do not do this unless you have to. Powering off the PI by unplugging it is not a safe practice.

# Raspberry PI Interface/commands information

When working with the Raspberry Pi, **do not unplug the power cable without shutting the system down first**. Doing so can corrupt the operating system or data.

## Shutting Down the Raspberry Pi

From the command line interface, run the following command to safely shut down the system:

```
Shell  
sudo shutdown -h now
```

This command will safely power down the Raspberry Pi. Once the **monitor screen goes completely black**, it is safe to disconnect the power.

## Restarting the Raspberry Pi

If you only need to restart the system, use this command:

```
Shell  
sudo shutdown -r now
```

## Scheduling a Shutdown or Restart

You can also schedule a shutdown or restart for a specific number of minutes by replacing **now** with the number of minutes.

```
Shell  
sudo shutdown -r 5
```

This command will **restart the Raspberry Pi in 5 minutes**.

## Finding the IP Address

If the IP address for the Bitfocus Companion dashboard changes (which is unlikely), you can find the current IP address by running:

```
Shell  
ifconfig
```

This command will display the available network interfaces and their IP addresses.

Because this system is configured to use **Ethernet only**, you should look for the **local Ethernet IP address**. Wi-Fi has been **disabled** on this device.

When reviewing the results, you may see:

- A **local IP address** (used to access the dashboard)
- A **loopback address** (used internally by the system)

Only the **local IP address** can be used to access the Companion dashboard. When connecting to the dashboard, remember to **include the Companion port number** in the address.

## Good to know Commands

| Name Of the command:                  | Description:                                       |
|---------------------------------------|----------------------------------------------------|
| <i>ifconfig</i>                       | Displays network interface configuration.          |
| <i>sudo shutdown -h now</i>           | This will shut the raspberry pi down, now.         |
| <i>sudo shutdown -h 5</i>             | This will shut down the raspberry Pi in 5 minutes. |
| <i>sudo shutdown -r now</i>           | This will restart the Pi, now                      |
| <i>sudo shutdown -r 5</i>             | This will restart the Pi in 5 minutes              |
| <i>sudo systemctl start companion</i> | Starts the Companion service                       |
| Name Of the command:                  | Description:                                       |

|                                         |                                                                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <i>sudo systemctl stop companion</i>    | Stop the Companion service                                                                                                            |
| <i>sudo systemctl restart companion</i> | Restarts the Companion service.                                                                                                       |
| <i>sudo systemctl status companion</i>  | Shows the current status of the Companion service.                                                                                    |
| <i>sudo companion-update</i>            | A custom executable that updates the Companion source code from GitHub.                                                               |
| <i>sudo apt update</i>                  | Synchronizes the list of packages on your system with the repositories.                                                               |
| <i>sudo raspi-config</i>                | Opens the configuration settings menu, where you can change the user password, set localization options, and configure other settings |
| <i>ip r</i>                             | Another command to reveal your Raspberry Pi's IP address                                                                              |

For more help see this Wiki on the official Bitfocus GitHub, [Here!](#)